Enhancing Post-Kidney Transplant Prognostication: An Interpretable Machine Learning Approach for Longitudinal Outcome Prediction

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Kidney transplantation offers life-extending treatment for patients with end-stage renal disease, yet long-term risks of graft loss and death persist. Traditional prediction models using only baseline data often fail to capture patients' evolving health status post-transplant. In this study, we propose a two-stage machine learning (ML) framework for dynamic, next-year risk prediction of graft loss and death, updated annually with newly available clinical and laboratory data. Using a multi-center cohort from the Swiss Transplant Cohort Study (STCS), we trained and evaluated five ML models across 13 years of follow-up, demonstrating that incorporating longitudinal data significantly improved predictive performance compared to baseline-only models. LightGBM achieved the strongest performance, with AUROC values up to 0.896 for graft loss and 0.797 for death. Our findings suggest that dynamic, interpretable ML models can enhance personalized risk stratification, offering a practical and scalable tool for guiding follow-up strategies and early interventions in kidney transplant recipients.

Introduction

Kidney transplantation is widely regarded as the optimal treatment for patients with end-stage renal disease, providing both extended survival and enhanced quality of life compared to long-term dialysis [1–3]. However, despite improvements in immunosuppression and surgical techniques, transplant recipients continue to face a substantial risk of graft loss and death in the post-transplant period [4]. While pre-transplant baseline factors such as donor type, immunological compatibility, and recipient comorbidities are well-established predictors of short- and long-term outcomes [5–7], the post-transplant course can deviate significantly from initial expectations due to complications

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such as acute rejection, infections, or other unforeseen events [8]. These post-transplant events can significantly influence patient outcomes and underscore the importance of ongoing risk assessment to guide timely clinical interventions.

In recent years, many machine learning-based approaches have been developed to estimate kidney transplant outcomes using baseline data alone [9–11]. Although these approaches have helped in initial decisions (for instance, transplant allocation or early post-operative care), they often do not capture the dynamic evolution of a recipient's health status. Similarly, the iBox risk prediction system [12], which is currently one of the best-known, validated risk prediction systems for kidney transplant outcomes, also provides a snapshot of risk at a single point in time rather than a continuously updated outlook. In particular, changes in renal function, the emergence of donor-specific antibodies, episodes of rejection, and additional comorbidities can all heavily influence long-term outcomes [13]. Neglecting these time-varying factors may lead to underestimation of risk in patients who develop complications or overestimation of risk in those who remain clinically stable, for example, in [14], the reported AUROC for predicting graft loss decreases from 0.97 to 0.70 as the prediction horizon increases from 1 year to 10 years using only the baseline characteristics. Consequently, a method that regularly updates risk estimates could be more responsive to each patient's evolving condition, thereby enhancing clinical decision-making and possibly improving graft longevity.

To address these challenges, we propose a continuous, two-stage ML model approach designed to predict next-year outcomes—specifically, graft loss or death—by iteratively integrating new post-transplant data on an annual basis. The first stage (Baseline Model) focuses on the initial year post-transplant, relying solely on pre-transplant data, which is typically the only information available for newly transplanted recipients. The second stage (Follow-up Model) then incorporates both baseline features and annual follow-up data, such as updated laboratory results, rejection episodes, or new comorbidities, to refine risk estimates for subsequent years. By continuously updating predictions, this approach captures the evolving clinical trajectory of each patient, offering a more nuanced and responsive alternative to static, baseline-only models. Importantly, this framework can both enhance individualized risk assessment in the clinic and serve as a flexible platform for integrating new diagnostic tools or therapies. Because the model updates predictions regularly, it could be adapted to rapidly assess the impact of novel biomarkers, point-of-care diagnostics, or therapeutic interventions on predicted outcomes—thereby accelerating clinical trials and observational studies in transplantation.

To the best of our knowledge, this is the first study to explore a continuous modeling framework for predicting kidney transplant outcomes over time. We present this work as a proof-of-concept, demonstrating how the integration of time-varying clinical data can enhance predictive accuracy and facilitate more personalized patient management. Our methodology employs explainable machine learning techniques to identify the most influential features in each prediction, providing insight into model drivers in clinical decision-making, such as adjustments to follow-up protocols. Additionally, we evaluate the model's performance across diverse sub-cohorts to assess its fairness and generalizability, aiming to ensure that this dynamic prediction system is applicable to a wide range of transplant recipients with varying risk profiles and backgrounds.

Results

Event Overview

We first examined the evolution of patient availability alongside the cumulative number of events (death or graft loss) across each post-transplant year. As depicted in Figure 1a for the death outcome, the size of the at-risk population decreases markedly over time, from over 4,000 in the first year to fewer than 100 by year 13. Simultaneously, the cumulative number of deaths increases gradually during the early years and more modestly in later years due to the shrinking at-risk population. However, it is important to note that the incidence proportion relative to the remaining at-risk population is higher in later years, indicating an elevated per-patient risk even as absolute event counts decline. A similar pattern emerges for graft loss, the at-risk population again shrinks substantially over the years. Notably, while the incidence proportion of graft loss is generally lower than the incidence proportion of death, the data underscore that long-term graft loss is still a significant concern for a subset of patients well beyond the immediate postoperative period. This highlights the temporal challenge in modeling outcomes over an extended timeline, as patients experience events (or become censored), the effective cohort diminishes, making it increasingly difficult to maintain predictive power in later years. Furthermore, the stacked histograms in Figure 1a illustrate that a substantial fraction of events occur within the first five years post-transplant, though many patients remain at risk far longer. These observations underscore both the time-dependent nature of graft loss or death and the importance of dynamic modeling approaches that can capture evolving risk profiles over multiple years.

Predictive Performance

Table 1 summarizes the predictive performance of the evaluated models, measured by the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC), for both Death and Graft Loss outcomes under the Baseline (Year 1) and Follow-up (Year 2+) schemes. Five machine learning models introduced in Section ML Prediction Models were assessed. Performance metrics were averaged across 5-fold nested cross-validation, with 95% confidence intervals provided to quantify uncertainty, and we specified how the 95% CIs were calculated in the Supplementary Information, Section 1.

The inclusion of follow-up data significantly enhances predictive performance compared to baseline-only models. For Death, AUROC values increase from approximately 0.60–0.70 at Baseline to above 0.75–0.80 in the Follow-up setting. Similarly, AUPRC values nearly double for certain models when time-varying post-transplant signals are incorporated, reflecting improved sensitivity to rare events. A comparable trend is observed for Graft Loss, where Baseline AUROCs are generally below 0.70 but exceed 0.80 in the follow-up phase. Among the evaluated models, LightGBM and TabPFN demonstrate the strongest overall performance. For Death, TabPFN achieves a marginally higher AUROC in both Baseline (0.689, 95% CI: 0.639–0.736) and Follow-up (0.802, 95% CI: 0.785–0.818) settings. However, LightGBM exhibits comparable discriminative ability and often outperforms in AUPRC, particularly for Graft Loss (0.276, 95% CI: 0.222–0.334 vs. 0.236, 95% CI: 0.186–0.292 for TabPFN). Given the low incidence proportion of adverse outcomes, AUPRC is a more critical metric as it reflects the model's ability to rank positive cases accurately. Consequently, LightGBM was selected for further analysis due to its robust performance, interpretability, and practical implementation advantages.

Figure 2 illustrates the evolution of LightGBM's predictive performance over post-transplant years. For Death, the AUROC starts at 0.64 in Year 1 and then rises sharply, reaching the mid to high 0.80s in later years. Similarly, the AUPRC nearly doubles from just over 0.04 in the first year to peak between 0.20 and 0.30 by Year 3 and beyond. These trends demonstrate the model's enhanced discriminative power as it leverages dynamic clinical data to better identify patients at elevated risk, despite the low incidence proportion of the outcome.

For Graft Loss, the AUROC begins at 0.68 in Year 1 and improves substantially, peaking $\sim$0.95 by Year 8. The AUPRC for Graft Loss follows a positive trajectory as well, increasing from around 0.08 in Year 1 to a high of 0.49 by Year 6. While some variability emerges in later years, likely due to a smaller cohort, the overall stability of these metrics underscores the value of continuous risk monitoring. In summary, LightGBM consistently captures evolving risk profiles over time, supporting timely and targeted clinical interventions.

Variable Importance

To gain insight into which factors drive the predictions, we computed SHAP values for each variable in the follow-up model. Figure 3 provides a comprehensive visualization for both the graft loss and death outcomes, structured into three distinct sub-panels per outcome: (1) **Beeswarm plot**: Depicts the distribution of SHAP values for the top 10 features, showing how each variable contributes positively or negatively to model predictions; (2) **Mean absolute SHAP values**: Displays the top 10 most influential features ranked by their absolute SHAP values, reflecting their overall impact on model predictions; (3) **SHAP-feature scatter plot**: Focuses on the most important feature, visualizing its relationship with the corresponding SHAP values.

For death prediction (Figure 3a), recipient age, eGFR, and cardiopulmonary disease indicator are the most impactful features. These variables align with clinical expectations, as older age and comorbidities significantly affect survival probabilities. Other relevant predictors include blood pressure levels, cholesterol measures, and metabolic health markers, suggesting that both baseline characteristics and follow-up signals shape the risk of death. The presence of "cardiopulmonary disease" and "other post-transplant diseases" among the top predictors reflects their role as composite indicators of severe post-transplant complications, capturing the cumulative burden of cardiovascular, pulmonary, and other major events. For graft loss prediction (Figure 3b), renal function indicators like eGFR, creatinine and proteinuria emerge as the dominant predictors, reinforcing the role of real-time kidney function in determining transplant viability. Additional contributors include donor age, cold ischemia time, and CKD progression markers, which highlight both recipient and donor factors in long-term graft survival.

These findings emphasize the importance of integrating both static pre-transplant variables and dynamic post-transplant updates to refine outcome predictions. Notably, our analysis indicates that the overall ranking of key variables remains stable over time, highlighting the persistent relevance of kidney function and systemic health markers. Further insights into how feature importance evolves annually are presented in Section 3 in the Supplementary Information. We also performed parallel SHAP analyses on the baseline model. Although baseline-only results identify many of the same broad categories of predictors (for example, donor characteristics and immunologic markers), the absence of real-time laboratory updates shifts some importance away from post-transplant factors. Furthermore, we examined the temporal dynamics of SHAP values year by year, focusing on how certain features might gain or lose relevance as time after transplant progresses. The results show that the overall rankings of top variables remain relatively stable. This analysis is also presented in Section 3 in the Supplementary Information.

Sub-cohort Analysis

We evaluated the performance across different patient subgroups during the follow-up period, presented in Figure 4. For death prediction, the model showed consistent AUROC values (0.72-0.89) across all subgroups, with lower but stable AUPRC values (0.07-0.20) reflecting the low event incidence proportion. Graft loss prediction demonstrated similar consistency in AUROC (0.85-0.90) with higher AUPRC values (0.25-0.40), which is in line with its higher performance in the overall evaluation. For donor-type specific analyses, we conducted targeted evaluations: "Living Donor: Related vs. Unrelated" analysis included only patients with living donor kidneys, while "Deceased Donor: DBD (donation after brain death) vs. DCD (donation after circulatory death)" analysis was performed only on deceased donor recipients. Both subgroups showed comparable predictive performance.

More granular analyses like cross-cohort validation (training on one subgroup and testing on another) and year-by-year breakdowns revealed the model's consistency over time and across different patient categories. These year-by-year plots show how AUROC, AUPRC, cohort size, and event incidence proportion evolve from the first post-transplant year through year 13, indicating that while performance remains robust, smaller samples in later years can introduce variability. Together, these findings demonstrate that our approach performs reliably across key subgroups and points in time. These results are detailed in Section 2 of the Supplementary Information, offering further insights into the model's generalizability and temporal stability.

Clinical Operating Point

A key step in translating machine learning model outputs into real-world practice is the decision of how to define high-risk patients. This involves choosing a specific threshold (or recall target) that meets clinical priorities, such as minimizing missed events or avoiding excessive false alarms. Table 2 illustrates how different recall levels affect metrics such as precision, specificity, and F1 score for both Death and Graft Loss in the follow-up model. This analysis is crucial because even a model with robust AUROC and AUPRC must be configured in a way that balances recall against precision in line with the care setting's resources and risk tolerance. The model predictions were post-hoc calibrated using isotonic regression.

For critical outcomes like death and graft loss, we looked at the high recall region to make sure the model captures most of the events. As an example, a recall target of 60% for Graft Loss yields a precision of 15.8%, but pushes specificity above 95%. Such a precision might be more acceptable in the clinical setting, yet under the extreme low-incidence proportion reality of late-year transplant outcomes, many false positives are inevitable. Similarly, a recall of 80% captures most at-risk patients but reduces precision to around 4.7%. This pattern underscores the difficulty of maintaining higher recall without incurring a large pool of false positives when the outcome is rare.

We have also provided a corresponding table for the Baseline Model in the first year post-transplant (see Supplementary Table 1), illustrating how similar operating-point considerations apply in the early stage. While its overall performance differs significantly from subsequent years, the same balance between recall and precision remains crucial for practical clinical decisions.

One way to reduce the impact of low incidence proportion is to extend the prediction window, thereby increasing the overall event rate. We explored 3-, 5-, and 10-year horizons (see Section 4 in the Supplementary Information), observing that precision improves considerably at similar recall levels once events are pooled over a longer timeframe. However, the main focus of our study remains on one-year prediction, which aligns with standard transplant follow-up practices. Ultimately, threshold selection should reflect each center's capacity and tolerance for false positives, reaffirming the need for flexible operating points in real-world deployment.

Discussion

In this study, we introduce a continuous, two-stage ML model approach that estimates next-year risks of graft loss or death in kidney transplant recipients by integrating newly available data on an annual basis. Our results indicate that incorporating time-updated clinical variables markedly enhances predictive accuracy compared to baseline-only approaches. This finding is consistent with the growing recognition that post-transplant outcomes are dynamically influenced by factors such as changes in renal function, rejection episodes, and emerging comorbidities [15, 16]. Accordingly, our approach provides a more adaptive model to inform immunosuppression management and the scheduling of targeted follow-up evaluations.

Our continuous two-stage framework demonstrates that the integration of time-updated features, such as estimated glomerular filtration rate (eGFR), creatinine, and proteinuria, substantially improves the accuracy of next-year outcome predictions relative to baseline-only models. Among the five candidate models evaluated, LightGBM exhibited superior performance, corroborating previous evidence of its efficacy in various clinical prediction tasks [17–19]. This enhancement is particularly evident during the intermediate post-transplant period when a patient's risk profile may evolve rapidly due to new comorbidities or rejection episodes. By systematically incorporating these annually updated indicators, the model offers a dynamic method for identifying patients at elevated risk, thereby enabling clinicians to adjust immunosuppressive regimens or schedule more frequent evaluations prior to the escalation of complications.

Our findings support the consensus that methods based solely on pre-transplant or immediate post-operative data may underestimate or overestimate patient risk in subsequent years after the transplant surgery [20, 21]. Although survival analysis incorporating time-varying covariates can capture temporal changes, they often require complex assumptions and may be challenging to interpret in routine clinical practice [22]. In contrast, our next-year prediction approach aligns with the established workflow at transplant centers, where patients typically undergo annual clinical assessments and laboratory testing. This renders our iterative methodology both practical and readily implementable, as new data can be incorporated into the follow-up model each year, providing a transparent, annually updated risk assessment to guide clinical decision-making.

Interpretability is a critical aspect of clinical predictive modeling, as clinicians need clear and justifiable explanations for model outputs before making changes to patient management strategies [23, 24]. To address this, we used SHAP value analysis to clarify the contributions of individual features, thereby enhancing transparency and building trust in the tool. For graft loss prediction, kidney function indicators like creatinine, eGFR, and proteinuria consistently ranked as the most influential predictors across years, underscoring the dominant role of real-time renal function in short-term risk assessment. In this context, the iBox risk prediction system [12] also identifies eGFR, proteinuria and antibody-mediated rejection as major contributors to graft survival. For death prediction, recipient age, proteinuria, and cardiopulmonary disease emerged as key predictors, reflecting the link between overall health status and organ function. Our temporal analysis (see Supplementary Figures 11-12) further shows that these feature rankings remain stable over time, highlighting how changes in renal function and comorbidities drive risk even beyond the early post-transplant phase. Additionally, the convergence of our top predictors with established clinical knowledge validates the model, as it did not rely on obscure or untested biomarkers but rather on factors already familiar to clinicians.

In terms of fairness, sub-cohort analyses showed that the model generally maintained strong discriminative performance across different patient categories, such as sex, donor type, and baseline risk factors. Although AUPRC varied with event incidence proportion in each subgroup, the performance trends remained relatively consistent. Some modest performance differences, however, warrant continued surveillance to ensure that no group is disproportionately misclassified. To address these potential imbalances, we recommend routine performance monitoring and, if necessary, recalibration of the model in specific cohorts.

Further insights emerge from the year-by-year breakdowns presented in Section 2 in the Supplementary Information, where each cohort's AUROC, AUPRC, sample size, and event incidence proportion are traced over thirteen years. These plots illustrate how performance can fluctuate due to increasing censoring rates and diminishing cohort sizes in the later years, highlighting both the strengths and limitations of our approach. The temporal stability we observe in most subgroups indicates that the model remains robust over an extended follow-up period, yet specific populations with smaller sample sizes may require targeted recalibration or complementary risk assessments. Additionally, our cross-cohort training and validation experiments, which are also detailed in Section 2 in the Supplementary Information, confirm that the model's predictive patterns generalize reasonably well, though certain subgroup combinations exhibit slightly lower metrics. This finding is not unexpected, given that training on one population does not always capture the nuanced clinical or demographic variations of another.

Taken together, these results reaffirm the value of a continuous, data-updating framework for predicting graft loss and death, while also emphasizing the need to regularly evaluate and fine-tune the model in different clinical settings. By incorporating both interpretability techniques and subcohort analyses, we aim to identify any systematic biases early and support equitable care for all transplant recipients.

Selecting an appropriate operating point for identifying high-risk patients is crucial for translating machine learning model predictions into actionable clinical decisions [25, 26]. In kidney transplant populations, the low incidence proportion of graft loss and death—especially in later post-transplant years—presents a significant challenge. As recall increases to capture more events, precision typically declines due to a rise in false positives. For instance, achieving 60% recall for graft loss results in a precision of 15.8% while maintaining specificity above 95%, a trade-off that may be acceptable when the priority is to minimize missed events. However, increasing recall to 80% further reduces precision to around 4.7%, underscoring the resource burden associated with such an approach in low-incidence proportion settings. A notable consideration in next-year prediction is the inflation of negative samples. For example, if a patient experiences an event in Year 10, all preceding years (Years 1–9) are treated as negative samples. While this approach expands the training dataset, it can skew the class distribution and dilute the predictive signal for rare events. Despite this limitation, next-year predictions also remain clinically relevant, as they align with the annual follow-up schedule and enable timely interventions.

As this is primarily a proof-of-concept study, these findings further underscore the challenges of achieving high precision under extremely imbalanced conditions, especially in the high-recall region where capturing most adverse events is critical. Nonetheless, demonstrating the feasibility of iteratively updating risk assessments each year provides a valuable foundation upon which more advanced modeling strategies or additional data inputs can be built to improve performance.

To mitigate the challenge of low precision, one strategy is to extend the prediction horizon by aggregating events over longer periods (e.g., 3, 5, or 10 years). This approach increases the event rate, thereby improving precision at comparable recall levels (see Section 4 in Supplementary Information). However, it shifts the focus from immediate, actionable next-year predictions to longer-term risk assessments, which are also clinically relevant in a clinical setting such as kidney transplantation where the clinical challenge with improving short term outcomes has shifted to extending long term graft survival. In practice, both short and long term outcomes are intertwined and as transplant centers often rely on yearly evaluations, making next-year risk assessments as well as longer term prediction a practical balance between timeliness and resource allocation [27, 28].

Ultimately, threshold selection must be tailored to local constraints, including available clinical resources, workflow capacity, and the acceptable risk of missing high-risk patients. Transplant centers may prioritize higher recall to minimize missed events or optimize precision to reduce false alarms, depending on their specific needs and risk tolerance. This flexibility ensures that the model can be adapted to diverse clinical workflows and patient populations, making threshold selection a vital yet customizable component of model deployment.

Despite the strengths of this approach, several limitations should be acknowledged. Although the STCS provides a rich, multi-center dataset, its modest cohort size and increasing attrition in later post-transplant years may introduce bias—particularly if patients lost to follow-up differ systematically from those retained. Our design focused on annual, instance-based predictions; however, alternative modeling paradigms such as survival analysis or multi-year forecasting may be more appropriate in certain clinical contexts. Furthermore, while multiple machine learning algorithms were evaluated, we did not exhaustively explore sequential or time-to-event modeling approaches.

Beyond these methodological considerations, external validation is a key priority for future work. While this study demonstrates feasibility and clinical utility within the STCS cohort, generalizability across diverse healthcare systems is essential for clinical deployment [29–31]. Although we reviewed large transplant registries (e.g., UNOS [32], ANZDATA [33], UKTR [34]), limitations in data granularity and accessibility placed them outside the immediate scope of this study. Future validation efforts should include these registries and, ideally, prospective clinical trials to assess whether model-informed decisions can improve patient outcomes. Integrating the framework into an interactive clinical decision-support tool may further enhance its utility by enabling real-time, patient-specific risk stratification.

In summary, our two-stage modeling framework offers a dynamic and interpretable alternative to static, baseline-only methods for predicting kidney transplant outcomes. By leveraging longitudinal data and machine learning, this approach supports more personalized immunosuppression strategies and earlier intervention for high-risk patients. Looking ahead, research should focus on scaling this framework to larger, more diverse populations, incorporating novel biomarkers, and integrating real-time clinical data to further refine and validate patient-specific transplant risk management.

Methods

Study Cohort

The study cohort was drawn from the Swiss Transplant Cohort Study (STCS) [35], a nationwide, prospective, multicenter cohort initiated in May 2008. The STCS includes comprehensive data from all six Swiss transplant centers and aims to monitor post-transplant outcomes systematically. By the end of 2019, the cohort comprised over 5,500 solid organ transplant recipients, with kidney transplants making up the majority. All solid organ transplant recipients (including islet transplantation) from the STCS were included, with enrollment at the time of transplantation (defined as reperfusion of the allograft). No exclusion criteria were applied. The data infrastructure integrates demographic, clinical, and genomic information, along with biobanking and patient-reported outcomes, to support translational research and personalized medicine.

The cohort provides a rich dataset for studying post-kidney transplant outcomes, including graft survival, patient survival, rejection episodes, and infectious diseases. Regular follow-ups occur at predefined intervals, and the STCS ensures high data granularity, capturing critical details such as donor-recipient matching, immunosuppressive therapies, longitudinal graft function assessments like estimated glomerular filtration rate (eGFR, which was calculated using the creatinine equation from [36]), and biopsy outcomes. This robust framework makes the STCS an invaluable resource for understanding and predicting post-transplant outcomes in kidney recipients. An overview of the characteristics of the kidney transplant patients is provided in Table 3. Ethical permission for the current study (project number FUP221) within the STCS was obtained from the Cantonal Ethics Committee of Zurich (BASEC-Nr.2024-00142).

Data Description

Our approach leverages two distinct sets of features: one set for the Baseline Model for pre-transplant period, and another set for the Follow-up Model (for years 1 and onward). A summary of these variables, organized by category, is shown in Table 4. We compiled 51 baseline predictors, which largely reflect pre-transplant conditions and patient characteristics. The predictors span five main categories: Demographics & History, Donor Type & Characteristics, Baseline Clinical & Lab values, Blood Group, and Immunologic & HLA. We then incorporate 47 time-updated features once the patient completes the first year without an event. These features are organized into four categories: Renal Function & Metabolics, Rejection & Histopathology, Infections & Comorbidities, and Chronic Kidney Disease (CKD) Staging. These post-transplant indicators capture the evolving clinical status of a patient, allowing the follow-up model to refine risk estimates each subsequent year.

ML Model Design

In this work, we propose a two-stage ML model approach to predict next-year outcomes (graft loss or death) on an annual basis. Specifically, we train: (1) **Baseline Model:** Predicts the risk of an adverse outcome during the first year post-transplant, using only pre-transplant data; (2) **Follow-up Model:** Predicts the risk of an adverse outcome during each subsequent year, leveraging both the original baseline features and any newly available data.

At each year t, a patient who has remained event-free up to that point is fed into the Follow-up Model along with their latest follow-up measurements, yielding a next-year risk estimate for year $t+1$. This process repeats until either the patient experiences the event or reaches the end of the follow-up period. A conceptual illustration of this two-stage approach is shown in Figure 1b. Importantly, we did not include the predicted risk score from the baseline or previous follow-up models as an additional predictor. Each year's prediction is based only on baseline features and newly available clinical data, ensuring that risk estimates remain interpretable and free from error propagation across years.

We opted for this annual, instance-based scheme rather than sequential or survival modeling approaches for the following reasons: (1) **Irregular Follow-up and Dropouts.** Our dataset includes patients who drop out or lose follow-up at various time points. Framing the problem as "next-year outcome" naturally accommodates different follow-up lengths, whereas recurrent neural network approaches [37, 38] often require consistent time-step intervals or specialized handling for missing sequences. (2) **Complexity.** Although sophisticated survival models (e.g., Cox regression with time-dependent covariates [39]) could be considered, such methods may introduce additional assumptions and complicate interpretability, as hazard ratios are inherently difficult to interpret and may not provide a clear picture of a covariate's effect [40]. Here, we primarily seek to demonstrate the feasibility of incorporating newly available data each year without imposing strict assumptions on the timing or structure of follow-up; (3) **Clinical Alignment.** In practice, transplant recipients are typically evaluated at regular (e.g., annual) intervals, making an 'annual instance' model both intuitive and easy to integrate into existing clinical workflows. For instance, after obtaining updated labs and examination results, clinicians can generate a one-year-ahead risk estimate to guide management decisions.

ML Prediction Models

We evaluated five machine learning approaches for the two-stage instance-based framework, comparing Logistic Regression (LR), Support Vector Machine (SVM), Multilayer Perceptron (MLP), LightGBM [41], and Tabular Prior-data Fitted Network (TabPFN) [42]. Among these, TabPFN is a state-of-the-art transformer-based foundation model pre-trained on millions of tabular datasets. It leverages in-context learning to adapt quickly to new tasks with minimal parameter tuning, making it particularly effective for small-to-medium datasets. TabPFN efficiently handles missing values, outliers, and categorical data, predicts in a single forward pass, and provides robust uncertainty quantification, which is especially valuable for clinical and other scientific applications [43]. Each model was applied in both Baseline and Follow-up settings to ensure consistent comparisons of predictive performance. We placed particular emphasis on TabPFN to investigate whether the current advanced foundation model concept offers significant advantages in predicting kidney transplant outcomes.

Evaluation Scheme

Given the relatively small size of our patient cohort, we implemented a 5-fold nested cross-validation (CV) strategy to rigorously evaluate model performance and mitigate the risk of overfitting. In each outer fold, the data were partitioned into training (70%), validation (10%), and test (20%) subsets. The inner folds were utilized for hyperparameter tuning and early stopping, where applicable, ensuring that model selection was performed on unseen data. The final 20% test subset was reserved for evaluation. Splitting was performed at the patient level: all patient-year instances from the same individual were consistently assigned to the same split (train, validation, or test) across folds. This ensured no leakage of patient-specific information between training and evaluation sets. This design yields less biased estimates of generalization error, which is particularly critical for small to moderate-sized clinical datasets. By aggregating performance metrics (e.g., AUROC, AUPRC) across all outer folds, we derived a more robust and reliable measure of each model's predictive capabilities. Further details regarding missing data handling and model development are provided in Section 1 in Supplementary Information. This evaluation framework not only reduces the risk of overfitting but also ensures a fair and comprehensive comparison of the five models described in Section ML Prediction Models.

Statistical Analysis

Following model development and evaluation, we conducted analyses to interpret, assess, and contextualize the predictive performance in a clinical setting. First, we applied SHAP (SHapley Additive exPlanations) [44] to quantify the contribution of each predictor to the model's output on an instance-by-instance basis. This approach provided transparent insights into which features, such as laboratory values, demographic factors, or comorbidity indicators, were most influential for individual predictions, thereby supporting interpretability and potential clinical validation of model decisions.

Next, to gauge robustness and fairness, we performed a sub-cohort analysis. Specifically, we stratified the testing cohort by relevant factors (e.g., sex, donor type, or baseline risk strata) and evaluated whether the discriminative ability of the model remained consistent across these subgroups. Any substantial drop in performance for a given subgroup could signal inequities or potential biases in how the model weights certain features, an important consideration for clinical adoption.

Finally, recognizing that clinicians typically require a specific balance between recall and precision, we explored clinical operating point selection to decide how best to convert the model's continuous risk scores into actionable decisions. By examining a range of thresholds and corresponding metrics, such as sensitivity, precision, positive predictive value, we identified practical cutoffs that could be used in different resource or risk-tolerance scenarios. This threshold selection stage is crucial because a purely algorithmic optimum may not align with the real-world imperative to minimize missed events in high-stakes healthcare settings, yet still keep the false-alarm burden manageable.

Data availability

The data analyzed in this study is subject to the following licenses/restrictions: because of the STCS data protection contract, data are not directly available but can be requested from the STCS Scientific Committee (FUP 221). Requests to access these datasets should be directed to https://www.stcs.ch/about/data-center.

Code availability

All scripts used for modelling and data analysis are available on GitHub: github.com/uzh-dqbm-cmi/Post-Transplant-Prognostication.

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Author contribution

B.F., M.S., J.N., and M.K. conceived the study. J.N. and M.K. supervised the study. L.F., M.K., M.Ko., C.V.D., A.L., D.G., J.V., T.S., D.S., S.S. and J.N. contributed to data acquisition. B.F. and M.S. performed the data cleaning, pre-processing, and label annotation. B.F. implemented the machine learning models and ran all experiments. B.F. performed the sub-cohort analysis, clinical operating point selection, and variable importance interpretation. M.S. and M.K. advised on modeling, statistical interpretation, and evaluation. Y.T., A.M., and J.N. advised on medical content and clinical relevance. B.F. wrote the manuscript with feedback and guidance from all other co-authors. All authors contributed to interpreting the findings and refining the manuscript.

Competing Interests

The authors declare no competing interests.

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References

[1] Hariharan, S., Johnson, C.P., Bresnahan, B.A., Taranto, S.E., McIntosh, M.J., Stablein, D.: Improved graft survival after renal transplantation in the united states, 1988 to 1996. New England Journal of Medicine **342**(9), 605–612 (2000)

[2] Wolfe, R.A., Roys, E., Merion, R.M.: Trends in organ donation and transplantation in the united states, 1999-2008. American Journal of Transplantation **10**(4), 961–972 (2010)

[3] Tonelli, M., Wiebe, N., Knoll, G., Bello, A., Browne, S., Jadhav, D., Klarenbach, S., Gill, J.: Systematic review: kidney transplantation compared with dialysis in clinically relevant outcomes. American journal of transplantation **11**(10), 2093–2109 (2011)

[4] Lamb, K., Lodhi, S., Meier-Kriesche, H.-U.: Long-term renal allograft survival in the united states: a critical reappraisal. American journal of transplantation **11**(3), 450–462 (2011)

[5] Van Loon, E., Bernards, J., Van Craenenbroeck, A.H., Naesens, M.: The causes of kidney allograft failure: more than alloimmunity. a viewpoint article. Transplantation **104**(2), 46–56 (2020)

[6] Morales, J.M., Marcén, R., Castillo, D., Andres, A., Gonzalez-Molina, M., Oppenheimer, F., Serón, D., Gil-Vernet, S., Lampreave, I., Gainza, F.J., *et al.*: Risk factors for graft loss and mortality after renal transplantation according to recipient age: a prospective multicentre study. Nephrology Dialysis Transplantation **27**(suppl_4), 39–46 (2012)

[7] Frischknecht, L., Deng, Y., Wehmeier, C., Rougemont, O., Villard, J., Ferrari-Lacraz, S., Golshayan, D., Gannage, M., Binet, I., Wirthmueller, U., *et al.*: The impact of pre-transplant donor specific antibodies on the outcome of kidney transplantation–data from the swiss transplant cohort study. Frontiers in immunology **13**, 1005790 (2022)

[8] Guedes, A.M., Malheiro, J., Fonseca, I., Martins, L.S., Pedroso, S., Almeida, M., Dias, L., Castro Henriques, A., Cabrita, A.: Over ten-year kidney graft survival determinants. International journal of nephrology **2012**(1), 302974 (2012)

[9] Badrouchi, S., Bacha, M.M., Ahmed, A., Ben Abdallah, T., Abderrahim, E.: Predicting long-term outcomes of kidney transplantation in the era of artificial intelligence. Scientific Reports **13**(1), 21273 (2023)

[10] Schwab, S., Sidler, D., Haidar, F., Kuhn, C., Schaub, S., Koller, M., Mellac, K., Stürzinger, U., Tischhauser, B., Binet, I., *et al.*: Clinical prediction model for prognosis in kidney transplant recipients (kidmo): study protocol. Diagnostic and prognostic research **7**(1), 6 (2023)

[11] Salaün, A., Knight, S., Wingfield, L., Zhu, T.: Predicting graft and patient outcomes following kidney transplantation using interpretable machine learning models. Scientific Reports **14**(1), 17356 (2024)

[12] Loupy, A., Aubert, O., Orandi, B.J., Naesens, M., Bouatou, Y., Raynaud, M., Divard, G., Jackson, A.M., Viglietti, D., Giral, M., et al.: Prediction system for risk of allograft loss in patients receiving kidney transplants: international derivation and validation study. Bmj **366** (2019)

[13] Loupy, A., Lefaucheur, C.: Antibody-mediated rejection of solid-organ allografts. New England Journal of Medicine **379**(12), 1150–1160 (2018)

[14] Yoo, K.D., Noh, J., Lee, H., Kim, D.K., Lim, C.S., Kim, Y.H., Lee, J.P., Kim, G., Kim, Y.S.: A machine learning approach using survival statistics to predict graft survival in kidney transplant recipients: a multicenter cohort study. Scientific reports **7**(1), 1–12 (2017)

[15] Sellarés, J., De Freitas, D., Mengel, M., Reeve, J., Einecke, G., Sis, B., Hidalgo, L., Famulski, K., Matas, A., Halloran, P.: Understanding the causes of kidney transplant failure: the dominant

role of antibody-mediated rejection and nonadherence. American Journal of Transplantation **12**(2), 388–399 (2012)

[16] Loupy, A., Hill, G.S., Jordan, S.C.: The impact of donor-specific anti-hla antibodies on late kidney allograft failure. Nature Reviews Nephrology **8**(6), 348–357 (2012)

[17] Lyu, X., Fan, B., Hüser, M., Hartout, P., Gumbsch, T., Faltys, M., Merz, T.M., Rätsch, G., Borgwardt, K.: An empirical study on kdigo-defined acute kidney injury prediction in the intensive care unit. Bioinformatics **40**(Supplement_1), 247–256 (2024)

[18] Hyland, S.L., Faltys, M., Hüser, M., Lyu, X., Gumbsch, T., Esteban, C., Bock, C., Horn, M., Moor, M., Rieck, B., *et al.*: Early prediction of circulatory failure in the intensive care unit using machine learning. Nature medicine **26**(3), 364–373 (2020)

[19] Yèche, H., Kuznetsova, R., Zimmermann, M., Hüser, M., Lyu, X., Faltys, M., Rätsch, G.: Hirid-icu-benchmark–a comprehensive machine learning benchmark on high-resolution icu data. arXiv preprint arXiv:2111.08536 (2021)

[20] Sprangers, B., Nair, V., Launay-Vacher, V., Riella, L.V., Jhaveri, K.D.: Risk factors associated with post–kidney transplant malignancies: an article from the cancer-kidney international network. Clinical Kidney Journal **11**(3), 315–329 (2018)

[21] Heldal, T.F., Åsberg, A., Ueland, T., Reisæter, A.V., Pischke, S.E., Mollnes, T.E., Aukrust, P., Reinholt, F., Hartmann, A., Heldal, K., *et al.*: Systemic inflammation early after kidney transplantation is associated with long-term graft loss: a cohort study. Frontiers in Immunology **14**, 1253991 (2023)

[22] Van Houwelingen, H., Putter, H.: Dynamic Prediction in Clinical Survival Analysis. CRC Press, Boca Raton (2011)

[23] Jin, W., Li, X., Fatehi, M., Hamarneh, G.: Guidelines and evaluation of clinical explainable ai in medical image analysis. Medical image analysis **84**, 102684 (2023)

[24] Liu, M., Ning, Y., Teixayavong, S., Mertens, M., Xu, J., Ting, D.S.W., Cheng, L.T.-E., Ong, J.C.L., Teo, Z.L., Tan, T.F., *et al.*: A translational perspective towards clinical ai fairness. NPJ Digital Medicine **6**(1), 172 (2023)

[25] Neves, M.R., Marsh, D.W.R.: Modelling the impact of ai for clinical decision support. In: Artificial Intelligence in Medicine: 17th Conference on Artificial Intelligence in Medicine, AIME 2019, Poznan, Poland, June 26–29, 2019, Proceedings 17, pp. 292–297 (2019). Springer

[26] Shin, H.J., Han, K., Son, N.-H., Kim, E.-K., Kim, M.J., Gatidis, S., Vasanawala, S.: Optimizing adult-oriented artificial intelligence for pediatric chest radiographs by adjusting operating points. Scientific Reports **14**(1), 31329 (2024)

[27] Renal Transplantation, E.E.G., *et al.*: European best practice guidelines for renal transplantation. section iv: Long-term management of the transplant recipient. iv. 10. pregnancy in renal transplant recipients. Nephrology, Dialysis, Transplantation: Official Publication of the European Dialysis and Transplant Association-European Renal Association **17**, 50–55 (2002)

[28] Group, K.D.I.G.O.K.T.W., *et al.*: Kdigo clinical practice guideline for the care of kidney transplant recipients. American journal of transplantation: official journal of the American Society of Transplantation and the American Society of Transplant Surgeons **9**, 1–155 (2009)

[29] Fan, B., Klatt, J., Moor, M.M., Daniels, L.A., Sanchez-Pinto, L.N., Agyeman, P.K., Schlapbach, L.J., Borgwardt, K.M.: Prediction of recovery from multiple organ dysfunction syndrome in pediatric sepsis patients. Bioinformatics **38**(Supplement_1), 101–108 (2022)

[30] Moor, M., Bennett, N., Plečko, D., Horn, M., Rieck, B., Meinshausen, N., Bühlmann, P., Borgwardt, K.: Predicting sepsis using deep learning across international sites: a retrospective

development and validation study. EClinicalMedicine **62** (2023)

[31] Moor, M., Rieck, B., Horn, M., Jutzeler, C.R., Borgwardt, K.: Early prediction of sepsis in the icu using machine learning: a systematic review. Frontiers in medicine **8**, 607952 (2021)

[32] Roberts, M.S., Angus, D.C., Bryce, C.L., Valenta, Z., Weissfeld, L.: Survival after liver transplantation in the united states: a disease-specific analysis of the unos database. Liver transplantation **10**(7), 886–897 (2004)

[33] McDonald, S.P., Russ, G.R.: Australian registries—anzdata and anzod. Transplantation Reviews **27**(2), 46–49 (2013)

[34] Byrne, C., Caskey, F., Castledine, C., Dawnay, A., Ford, D., Fraser, S., Williams, A.: Uk renal registry. Nephron **139**(1), 1–12 (2018)

[35] Stampf, S., Mueller, N.J., Delden, C., Pascual, M., Manuel, O., Banz, V., Binet, I., De Geest, S., Bochud, P.-Y., Leichtle, A., *et al.*: Cohort profile: The swiss transplant cohort study (stcs): A nationwide longitudinal cohort study of all solid organ recipients in switzerland. BMJ open **11**(12), 051176 (2021)

[36] Levey, A.S., Stevens, L.A., Schmid, C.H., Zhang, Y., Castro III, A.F., Feldman, H.I., Kusek, J.W., Eggers, P., Van Lente, F., Greene, T., *et al.*: A new equation to estimate glomerular filtration rate. Annals of internal medicine **150**(9), 604–612 (2009)

[37] Lipton, Z.: Learning to diagnose with lstm recurrent neural networks. arXiv preprint arXiv:1511.03677 (2015)

[38] Choi, E., Schuetz, A., Stewart, W.F., Sun, J.: Using recurrent neural network models for early detection of heart failure onset. Journal of the American Medical Informatics Association **24**(2), 361–370 (2017)

[39] Zhang, Z., Reinikainen, J., Adeleke, K.A., Pieterse, M.E., Groothuis-Oudshoorn, C.G.: Time-varying covariates and coefficients in cox regression models. Annals of translational medicine **6**(7) (2018)

[40] Bellera, C.A., MacGrogan, G., Debled, M., De Lara, C.T., Brouste, V., Mathoulin-Pélissier, S.: Variables with time-varying effects and the cox model: some statistical concepts illustrated with a prognostic factor study in breast cancer. BMC medical research methodology **10**(1), 20 (2010)

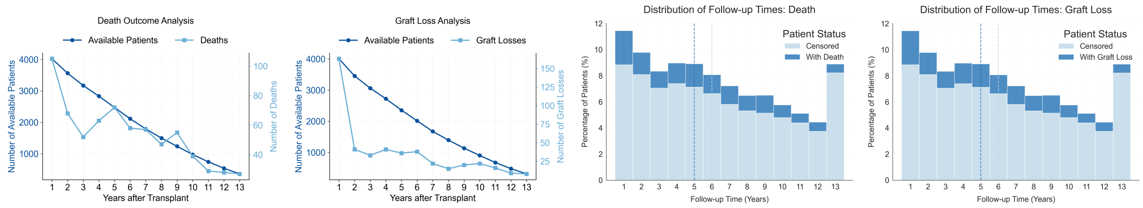
[41] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: Lightgbm: A highly efficient gradient boosting decision tree. Advances in neural information processing systems **30** (2017)

[42] Hollmann, N., Müller, S., Eggensperger, K., Hutter, F.: Tabpfn: A transformer that solves small tabular classification problems in a second. arXiv preprint arXiv:2207.01848 (2022)

[43] Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S.B., Schirrmeister, R.T., Hutter, F.: Accurate predictions on small data with a tabular foundation model. Nature **637**(8045), 319–326 (2025)

[44] Lundberg, S.: A unified approach to interpreting model predictions. arXiv preprint arXiv:1705.07874 (2017)

(a) **Event distribution**



(b) **Two-stage modeling**

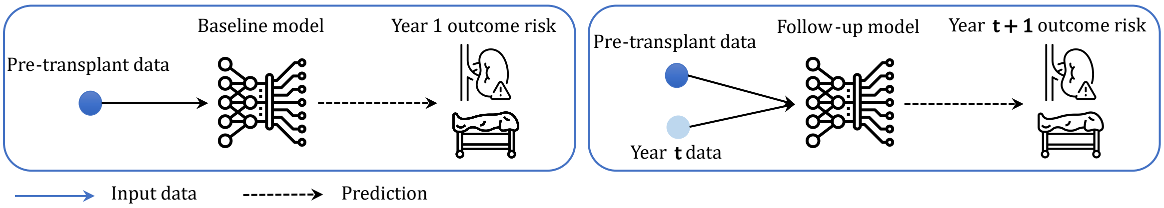


Fig. 1: **Overview of the event distribution and two-stage modeling approach.** **(a)** Event distribution. The first two plots show, for death and graft loss, the number of at-risk patients over successive post-transplant years (left y-axis) together with the number of events observed in each year (right y-axis). The stacked histograms depict the distribution of follow-up lengths: each x-axis bin corresponds to the post-transplant year in which a patient was last observed (years 1–13). Bars are divided into patients who remained event-free (censored; light blue) and those who experienced the event (dark blue); the total column height therefore represents the proportion of the cohort whose follow-up ended in that year because of either censoring or the event. A substantial fraction of events occur within the first five years, although many patients remain at risk much longer. **(b)** Two-stage modeling. (Left) The Baseline model uses only pre-transplant data to estimate the risk of an outcome during the first post-transplant year. (Right) For every subsequent event-free year t ($t = 1, 2, \ldots, 12$), the Follow-up model uses the baseline data plus the follow-up measurements collected during year t ("Year t data") to estimate the risk of an outcome in year $t+1$ ("Year $t+1$ outcome risk"). Thus, the first follow-up prediction occurs at the end of Year 1 (using Year 1 data to predict Year 2 risk), and the process repeats annually until an event or censoring.

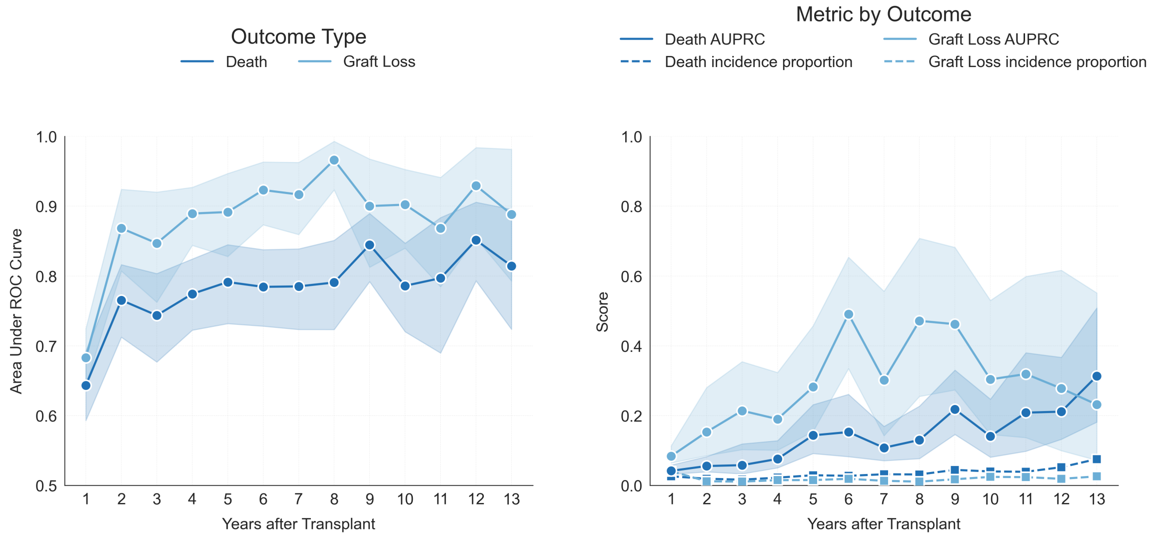
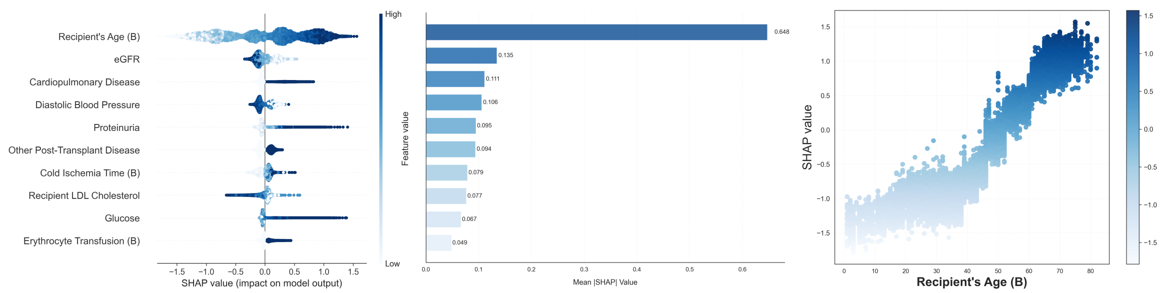


Fig. 2: **Year-by-year performance of the LightGBM model.** Shown for (Left) AUROC and (Right) AUPRC. Light blue points represent Graft Loss, while dark blue points represent Death. In the AUPRC plot, the incidence proportion of the events is also visualized using dashed lines under the curves. The shaded area indicates the 95% confidence intervals to quantify the uncertainty of the prediction.

(a) **SHAP value - Death**



(b) **SHAP value - Graft loss**

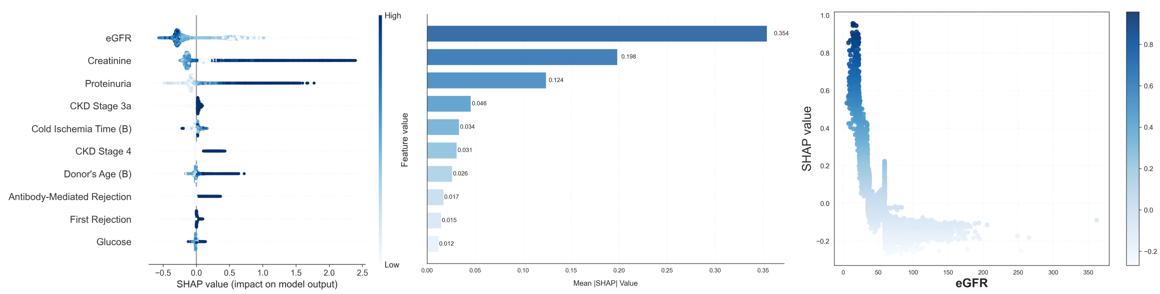
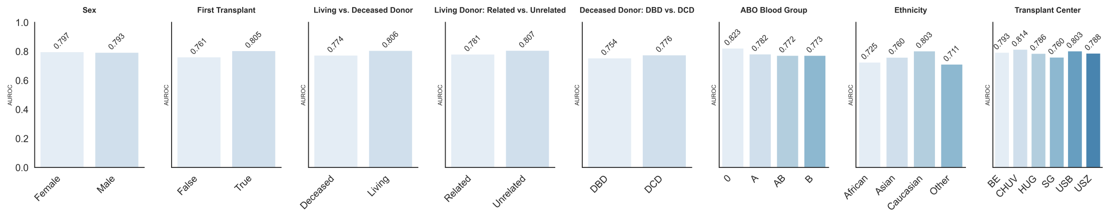
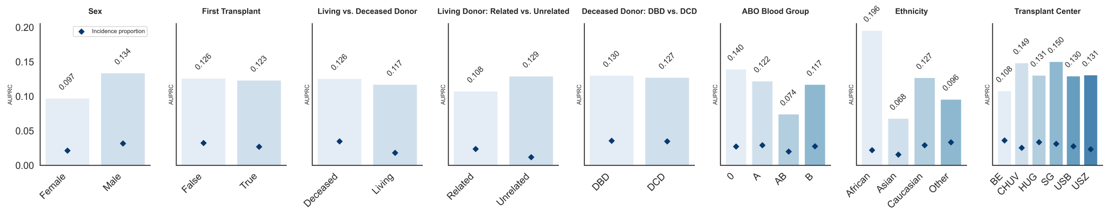


Fig. 3: **SHAP value analysis for the follow-up model predictions.** Each outcome type is visualized using three panels: (Left) SHAP beeswarm plots illustrating feature impact distributions of the top 10 features. The (B) after the feature name indicates a baseline characteristic. High feature values are depicted in dark blue, low values in lighter blue. (Middle) the top 10 features ranked by mean absolute SHAP values. (Right) 2D scatter plots showing the relationship indicates a baseline characteristic between SHAP values and actual feature values for the most important predictor. Panel (a) corresponds to the death prediction and Panel (b) corresponds to the graft loss prediction. These visualizations provide interpretability by highlighting how individual features influence the model predictions.

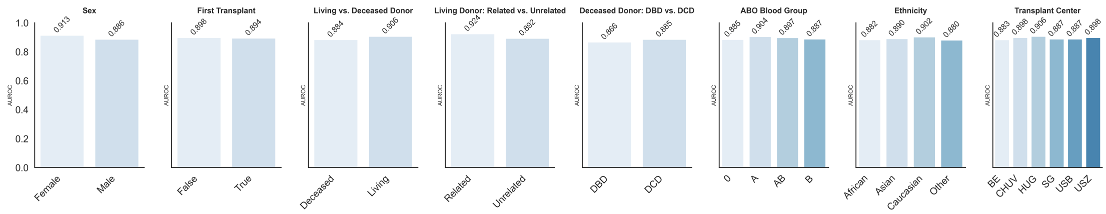
(a) **Death - AUROC**



(b) **Death - AUPRC**



(c) **Graft Loss - AUROC**



(d) **Graft Loss - AUPRC**

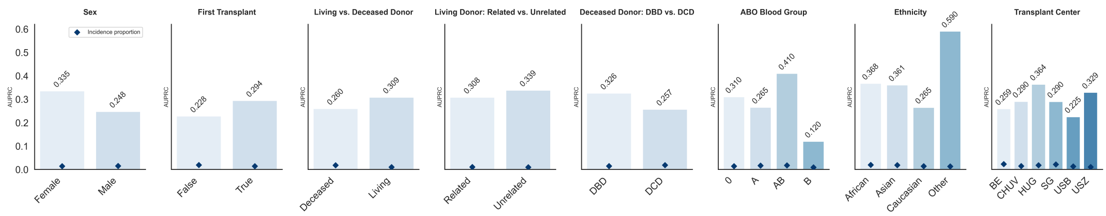


Fig. 4: **Subcohort analysis for the follow-up model.** Stratified by variables such as sex, donor type, blood group, ethnicity, and transplant center. The plots show performance for Death (top two rows) and Graft Loss (bottom two rows) outcomes, measured by both AUROC (first and third rows) and AUPRC (second and fourth rows). Bars indicate the performance metric values in each subgroup, while the diamonds in the AUPRC plots represent the incidence proportion of the outcome within that subgroup. For donor-type specific analyses, "Living Donor: Related vs. Unrelated" includes only patients with living donor kidneys, while "Deceased Donor: DBD vs. DCD" includes only deceased donor recipients.

Model	Death				Graft Loss			
	Baseline		Follow-up		Baseline		Follow-up	
	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
LR	0.620 (0.567-0.669)	0.036 (0.027-0.046)	0.774 (0.756-0.792)	0.097 (0.082-0.113)	0.596 (0.548-0.641)	0.067 (0.049-0.093)	0.859 (0.837-0.880)	0.225 (0.178-0.279)
SVM	0.653 (0.604-0.696)	0.042 (0.031-0.054)	0.774 (0.761-0.786)	0.105 (0.090-0.121)	0.644 (0.610-0.689)	0.056 (0.044-0.073)	0.837 (0.813-0.860)	0.210 (0.175-0.242)
MLP	0.641 (0.595-0.689)	**0.050** **(0.033-0.067)**	0.633 (0.609-0.658)	0.096 (0.088-0.107)	0.588 (0.542-0.631)	0.059 (0.046-0.075)	0.809 (0.779-0.838)	0.184 (0.139-0.236)
LGBM	0.644 (0.593-0.693)	0.041 (0.030-0.054)	0.797 (0.779-0.814)	0.122 (0.104-0.142)	**0.684** **(0.644-0.725)**	**0.084** **(0.062-0.108)**	**0.896** **(0.875-0.914)**	**0.276** **(0.222-0.334)**
TabPFN	**0.689** **(0.639-0.736)**	0.048 (0.035-0.061)	**0.802** **(0.785-0.818)**	**0.124** **(0.104-0.146)**	0.664 (0.627-0.703)	0.067 (0.053-0.083)	0.857 (0.836-0.877)	0.236 (0.186-0.292)

Incidence proportions:
Baseline Death: 2.61%, Follow-up Death: 2.79%, Baseline Graft Loss: 4.07%, Follow-up Graft Loss: 1.49%

Table 1: **Performance comparison of machine learning models.** The prediction performance is presented for the Death and Graft Loss outcomes at baseline and follow-up timepoints. The table reports Area Under the Receiver Operating Characteristic curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC) with 95% confidence intervals shown in parentheses. Best performing metrics in each category are shown in bold.

	Death (Follow-up)					Graft Loss (Follow-up)				
Recall	**Threshold**	**Precision**	**Specificity**	**Balanced Acc.**	**F1**	**Threshold**	**Precision**	**Specificity**	**Balanced Acc**	**F1**
0.5	0.039	0.104	0.873	0.693	0.173	0.078	0.228	0.974	0.743	0.315
0.6	0.027	0.083	0.807	0.707	0.146	0.056	0.158	0.951	0.776	0.251
0.7	0.012	0.071	0.735	0.719	0.129	0.020	0.097	0.900	0.805	0.171
0.8	0.008	0.058	0.625	0.716	0.109	0.006	0.047	0.751	0.784	0.090
0.9	0.003	0.048	0.487	0.696	0.092	0.005	0.039	0.659	0.786	0.075

Table 2: **Clinical operating-points analysis.** Done at different recall targets based on calibrated predictions (using isotonic regression) for death and graft loss in the follow-up period. Reported metrics include precision, specificity, balanced accuracy, and F1 score, evaluated at recall levels from 50% to 90%.

Variable	**Missing**	**Overall**	**BE** *Bern*	**CHUV** *Lausanne*	**HUG** *Geneva*	**SG** *St. Gallen*	**USB** *Basel*	**USZ** *Zurich*
n	-	4728	721	790	485	306	1118	1308
Recipient Age, Mean (std)	0	51.4 (15.7)	51.2 (17.0)	52.0 (16.1)	52.8 (15.0)	52.4 (13.4)	52.7 (13.8)	49.2 (16.7)
Donor Age, Mean (std)	36	52.4 (16.1)	54.3 (15.4)	51.9 (15.0)	52.8 (15.6)	55.5 (14.8)	52.6 (17.5)	50.6 (16.0)
Sex (Recipient), n (%)	0							
Female		1677 (35.5)	258 (35.8)	258 (32.7)	179 (36.9)	100 (32.7)	383 (34.3)	499 (38.1)
Male		3051 (64.5)	463 (64.2)	532 (67.3)	306 (63.1)	206 (67.3)	735 (65.7)	809 (61.9)
Sex (Donor), n (%)	40							
Female		2331 (49.7)	359 (49.9)	418 (53.3)	242 (50.2)	137 (44.8)	567 (51.1)	608 (47.3)
Male		2357 (50.3)	361 (50.1)	366 (46.7)	240 (49.8)	169 (55.2)	543 (48.9)	678 (52.7)
s **Donation Type, n (%)**	0							
Deceased		2983 (63.1)	474 (65.7)	462 (58.5)	281 (57.9)	217 (70.9)	632 (56.5)	917 (70.1)
Living		1745 (36.9)	247 (34.3)	328 (41.5)	204 (42.1)	89 (29.1)	486 (43.5)	391 (29.9)
First Transplant, n (%)	0							
No		753 (15.9)	121 (16.8)	130 (16.5)	75 (15.5)	45 (14.7)	164 (14.7)	218 (16.7)
Yes		3975 (84.1)	600 (83.2)	660 (83.5)	410 (84.5)	261 (85.3)	954 (85.3)	1090 (83.3)

Table 3: Overview of STCS kidney transplant recipient and donor characteristics.

Model	Category	# Vars	Example Variables
Baseline	Demographics & History	6	Age of Recipient, Prev. Transplant
	Donor Type & Characteristics	4	Donor Age, Living vs. Deceased Donor
	Baseline Clinical/Lab	9	Baseline eGFR, Baseline Glucose
	Blood Group	2	Recipient Blood Group, Donor Blood Group
	Immunologic & HLA	30	DSA Class, HLA typing for donor and recipient
Follow-Up	Renal Function & Metabolics	9	eGFR, Blood Pressure
	Rejection & Histopathology	24	First Rejection, Antibody-Mediated Rejection
	Infections & Comorbidities	9	Cardiopulmonary Disease, Virus Infection
	CKD Staging	5	CKD Stage 2, Stage 3a, Stage 4

Table 4: Overview of Predictor Variables for Baseline and Follow-up Models. The Baseline Model predicts the outcome for the first post-transplant year using only pre-transplant data, while the Follow-up Model predicts the outcome for the next year (Year $t + 1$) using pre-transplant data combined with follow-up measurements from Year t ($t = 1, 2, \dots$).